SPACEX FALCON 9

First-Stage Landing Prediction

IBM Applied Data Science Capstone Project

Data collection • SQL • EDA • Interactive visualization • Classification

GitHub: github.com/Yossefassem79/SpaceX-Falcon-9-Landing-Prediction

Introduction

Project context, prediction target, and scope

BACKGROUND

SpaceX Falcon 9 first stages may land on a drone ship, a ground pad, or the ocean, or fail to land successfully. Landing success is an important part of rocket reusability and launch-cost analysis.

PROJECT QUESTION

Can historical launch characteristics help predict first-stage landing success?

The project explores launch records, payload mass, orbit, flight number, and launch site, then compares four classification algorithms. The analysis identifies associations in the dataset; it does not prove causation.

Executive Summary

A data-driven study of Falcon 9 first-stage landing outcomes

PROJECT GOAL

Explore launch records and build a classification model to predict whether a Falcon 9 first stage lands successfully.

KEY TAKEAWAYS

- Historical landing success generally improved over time.
- Landing outcomes vary across orbits and launch contexts.
- Logistic Regression was selected by the notebook's score comparison at 83.33% accuracy.
- The model scores shown for all four classifiers were equal, so this is a tie—not evidence of unique superiority.

Problem Statement & Objectives

Why predict first-stage landing success?

CHALLENGE

Reusable launch vehicles can reduce launch costs. Understanding landing outcomes can support better mission planning.

OBJECTIVES

1. Collect and prepare launch records
2. Explore patterns using plots and SQL
3. Map launch sites and nearby features
4. Build and compare classification models
5. Communicate findings with an interactive dashboard

Project Workflow

End-to-end applied data science process

01

Collect

API / web scraping

→

02

Wrangle

Clean and label data

→

03

Explore

EDA + SQL

→

04

Visualize

Folium + Dash

→

05

Model

Train and evaluate

Data Collection & Wrangling

Preparing launch records for analysis

DATA COLLECTION

- Launch records assembled from course-provided data sources
- Web scraping used for relevant launch information
- SQL data stored in a local SQLite database for query-based analysis

WRANGLING & FEATURES

- Standardized key fields and prepared dates
- Used landing outcome to define a binary class label
- Class 1 represents successful landing; Class 0 represents unsuccessful / non-success outcomes according to the project labeling

Note: detailed cleaning decisions should be interpreted alongside the notebook and dataset.

EDA with SQL: Launch Sites & Milestones

Using SQL to answer targeted questions

Task 1

Display the names of the unique launch sites in the space mission

```
%sql select distinct launch_site from SPACEXTABLE ;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Four unique launch sites were returned by SELECT DISTINCT.

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql select * from SPACEXTABLE where launch_site like "CCA%" limit 5 ;
```

Python

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The earliest successful ground-pad landing in the query result was 2015-12-22.

EDA with SQL: Outcome Counts

Grouped counts and date-based filtering

Task 5

List the date when the first succesful landing outcome in ground pad was acheiv

Hint:Use min function

```
%sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

The raw mission_outcome field contains inconsistent Success labels, which should be normalized before aggregation.

Task 9

List the records which will display the month names, failure landing_outc
for the months in year 2015.

**Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2
substr(Date,0,5)='2015' for year.**

```
%sql SELECT substr(Date, 6, 2) AS Month, Landing_Outcome,Booster_Version,Launch
```

```
 * sqlite:///my_data1.db
Done.
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

For 2010-06-04 to 2017-03-20, No attempt was the most frequent landing outcome in this result (10).

EDA with SQL: Monthly Failure Records

Date substring extraction in SQLite

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT mission_outcome, COUNT(*) AS total FROM SPACEXTABLE GROUP BY mission_outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

The displayed 2015 records identify failures (drone ship) in:

- **January (01)**
- **April (04)**

EDA & Data Visualization Methodology

How the exploratory analysis was performed

DATA EXPLORATION

- Inspect launch records and relevant columns.
- Compare the binary landing class (0 / 1) across launch sites, orbit, payload mass, and flight number.
- Group records by year and orbit, then calculate the mean of Class as a descriptive success rate.

VISUAL ENCODING

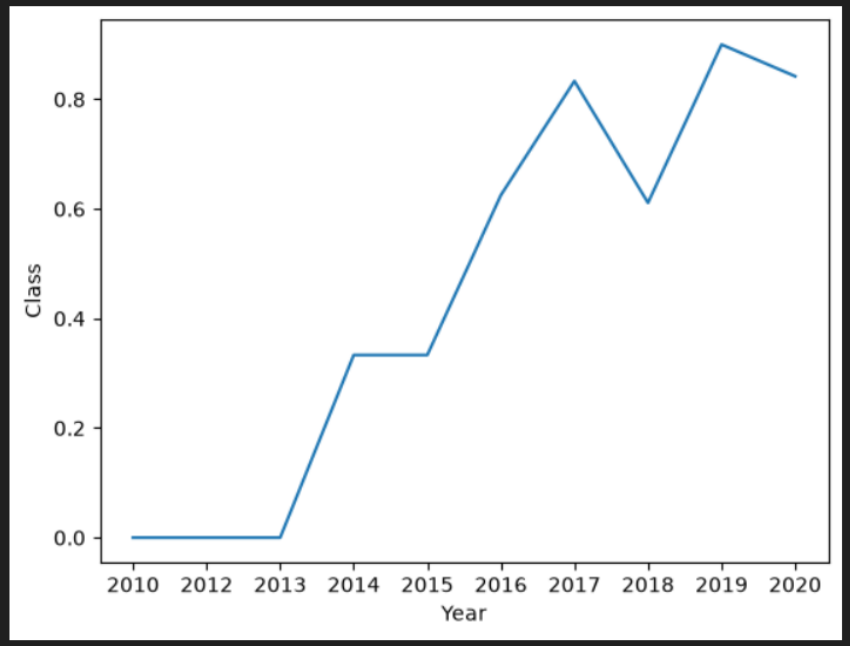
- Line chart: yearly landing success rate.
- Bar chart: mean success rate by orbit.
- Class-colored scatter plots: payload mass / flight number versus launch site or orbit.

Rates for categories with few launches should be interpreted cautiously.

EDA: Landing Success Over Time

Yearly mean of the binary landing class

```
# Plot a line chart with x axis to be the extracted year and y axis to be the success rate
yearly_success = df.groupby('Date')['Class'].mean().reset_index()
sns.lineplot(x = "Date", y = "Class", data = yearly_success)
plt.xlabel("Success Rate")
plt.xlabel("Year")
plt.show()
```



OBSERVATION

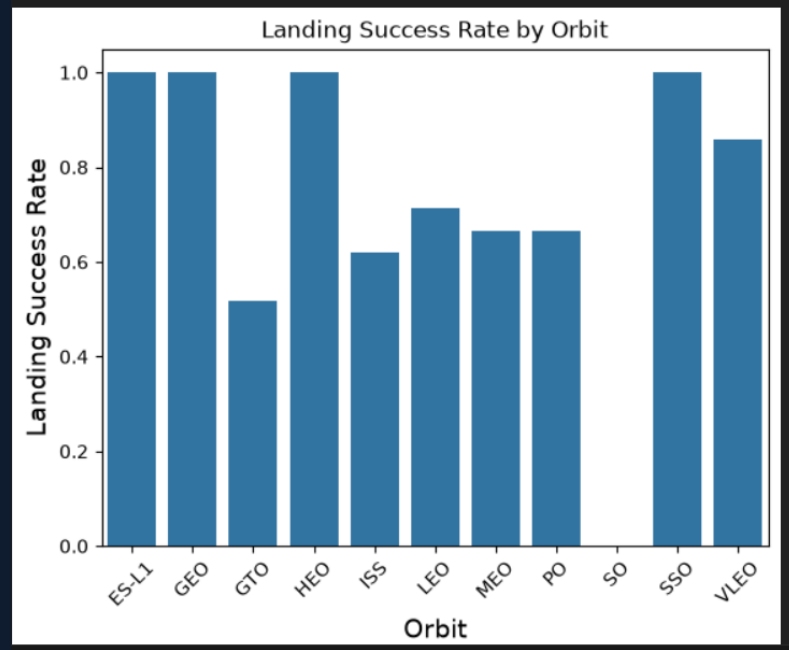
- Success rose from 0 in the earliest years shown to over 0.8 in 2017 and 2019.
- The rate dipped in 2018 and slightly in 2020.
- Yearly rates can fluctuate because the number and type of launches vary.

EDA: Landing Success by Orbit

Comparing average success rates across orbit categories

```
# HINT use groupby method on Orbit column and get the mean of Class column
orbit_success = df.groupby('Orbit')['Class'].mean()
sns.barplot(x=orbit_success.index, y=orbit_success.values)

plt.xlabel('Orbit', fontsize=14)
plt.ylabel('Landing Success Rate', fontsize=14)
plt.title('Landing Success Rate by Orbit')
plt.xticks(rotation=45)
plt.show()
```



KEY INSIGHT

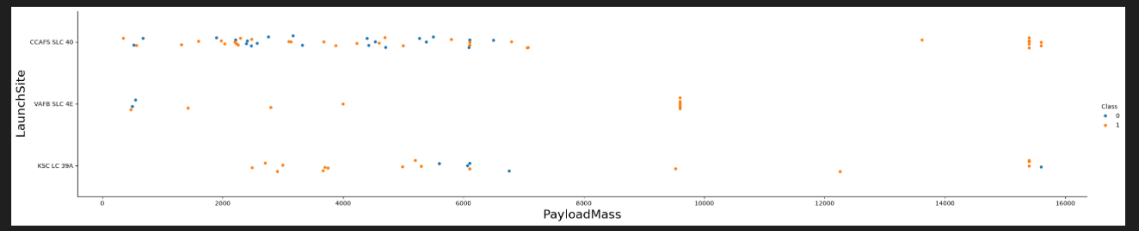
- GTO is about 0.52 and ISS about 0.62 in the shown plot.
- Some low-frequency orbit categories show 100% success.
- Small samples can make extreme rates unstable; interpret rare-orbit values cautiously.

EDA: Payload, Flight Number & Launch Site

Inspecting distributions of successes and failures

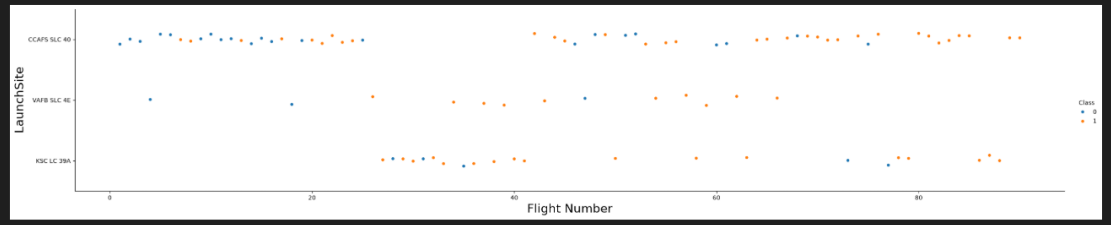
```python
# Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
sns.catplot(x = 'PayloadMass',y = 'LaunchSite', data = df , hue= 'Class',aspect= 5)
plt.xlabel("PayloadMass",fontsize=20)
plt.ylabel("LaunchSite",fontsize=20)
plt.show()
```

Python



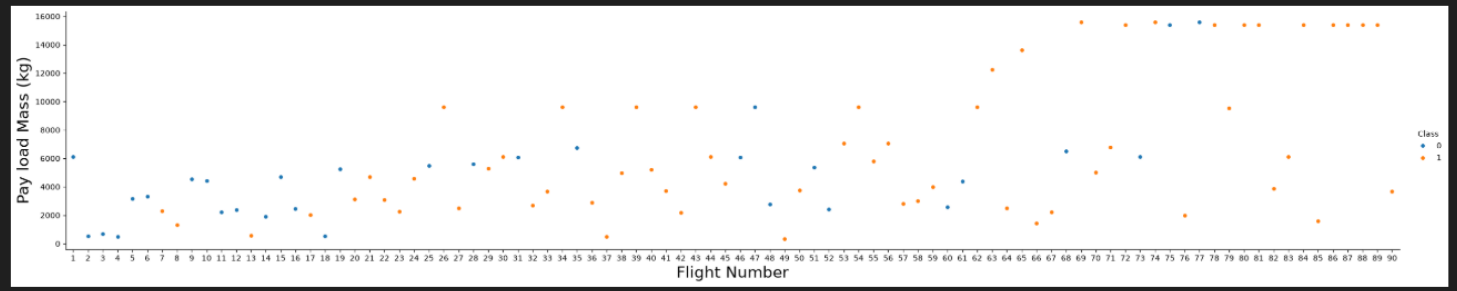

```python
# Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(x = 'FlightNumber',y = 'LaunchSite', data = df , hue= 'Class',aspect= 5)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("LaunchSite",fontsize=20)
plt.show()
```

Python




```python
sns.catplot(y="PayloadMass", x="FlightNumber", hue="Class", data=df, aspect = 5)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("Pay load Mass (kg)",fontsize=20)
plt.show()
```

Python

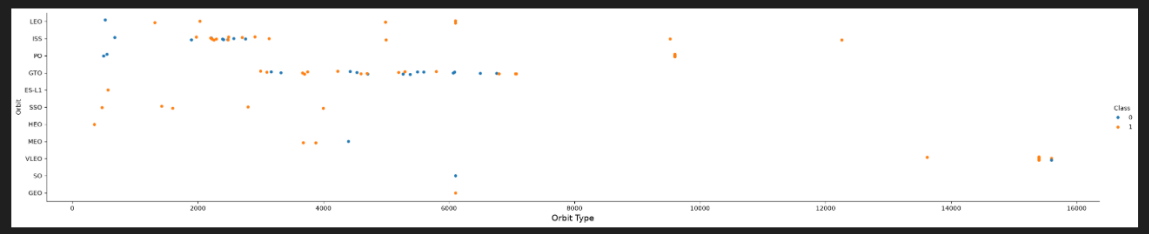


EDA: Flight and Payload Patterns by Orbit

Class-colored scatter plots

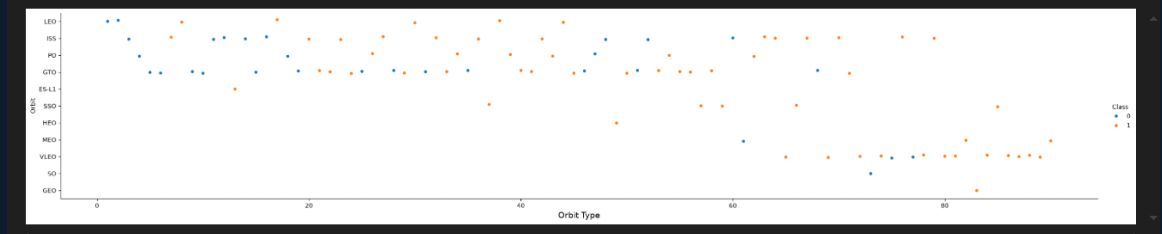
```
Click to add a breakpoint
int chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
sns.catplot(x= "PayloadMass", y = "Orbit", data = df ,hue = "Class", aspect = 5)
plt.xlabel("Payload Mass",fontsize= 14)
plt.xlabel("Orbit Type",fontsize= 14)
plt.show()
```

Python




```
# Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.catplot(x= "FlightNumber", y = "Orbit", data = df ,hue = "Class", aspect = 5)
plt.xlabel("Flight Number",fontsize= 14)
plt.xlabel("Orbit Type",fontsize= 14)
plt.show()
```

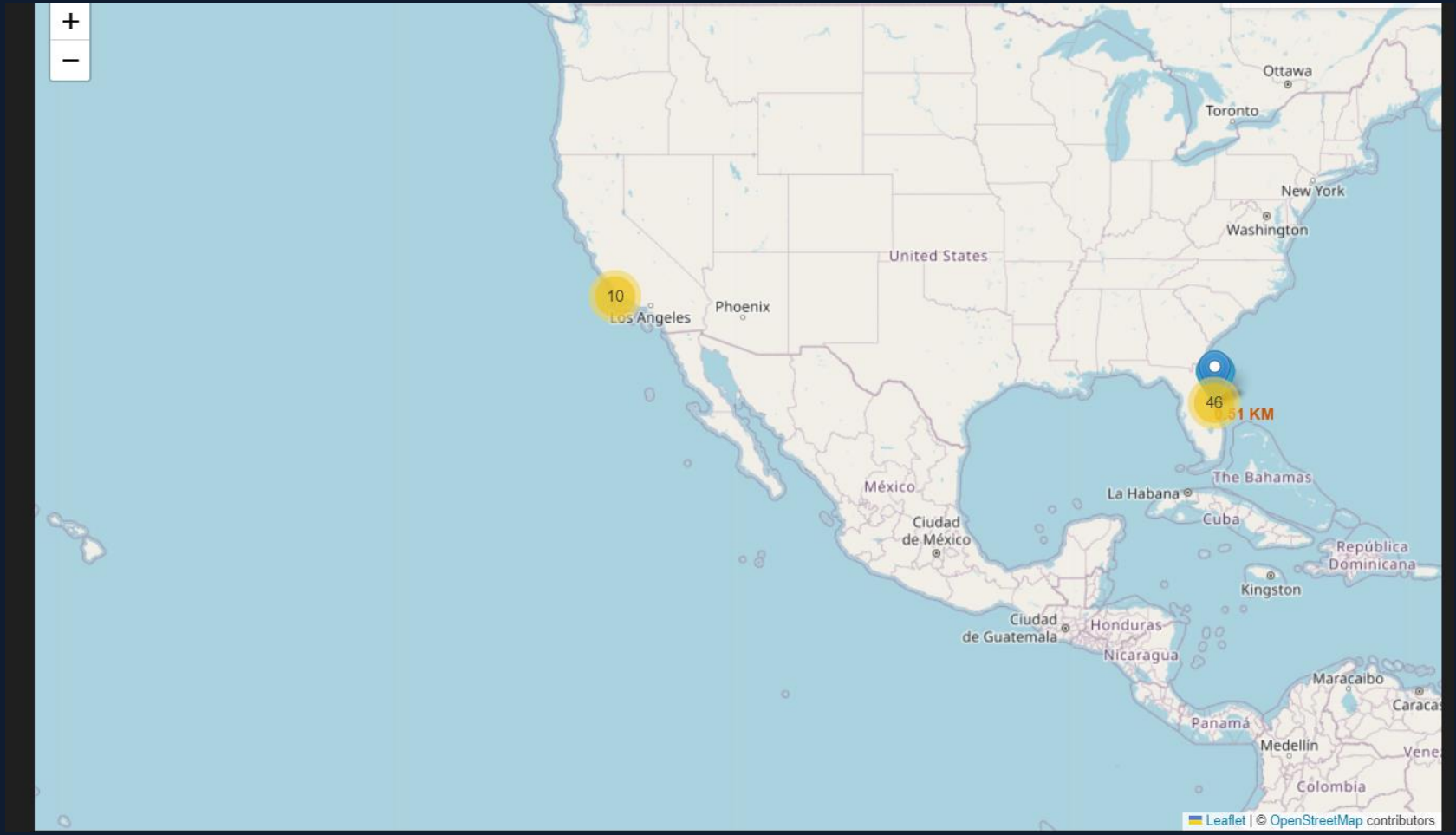
Python



Class 0 and Class 1 are shown in different colors to compare landing outcomes across flight number / payload mass and orbit categories.

Interactive Mapping with Folium

Geographic view of launch-related locations



MAP FEATURES

- Leaflet / OpenStreetMap base map
- Clustered markers for locations
- A displayed distance annotation near the Florida launch area
- Zoom and pan for geographic context

The submitted screenshot is zoomed out; individual launch-site details are easier to inspect by zooming in.

Interactive Analytics with Plotly Dash

Dashboard for exploring launch records



CORE INTERACTIONS

- Filter by launch site
- Explore successful launches by site
- Adjust payload mass range
- Compare payload mass with landing success

The diagram summarizes the previously shared dashboard components; it is not a new screenshot of the running app.

Predictive Analysis Methodology

Classification pipeline and model comparison

MODELS EVALUATED

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)

The notebook also uses GridSearchCV to explore candidate hyperparameters.

EVALUATION

The models are compared using test accuracy and confusion matrices. Accuracy is the proportion of correct predictions, but it does not show which error type is most common.

The displayed test accuracy is 83.33% for all four models. Therefore, Logistic Regression is the first model returned by the max-score code, not a uniquely higher-scoring model.

Predictive Analysis: Classification

Comparing Logistic Regression, SVM, Decision Tree, and KNN

TASK 12

Find the method performs best:

```
scores = {
    "Logistic Regression": logreg_score,
    "SVM": SVM_score,
    "Decision Tree": tree_score,
    "KNN": knn_score
}

best_model = max(scores, key=scores.get)
print(f"The best performing method is: {best_model} with a score of {scores[best_model]:.2%}")
```

```
The best performing method is: Logistic Regression with a score of 83.33%
```

REPORTED TEST ACCURACY

83.33%

All four models were shown with the same test accuracy in the notebook. The max-score code returns Logistic Regression because it appears first in the dictionary when the scores tie.

Accuracy alone is not enough: use confusion matrices and precision/recall/F1 to assess error types.

Key Insights & Limitations

What the exploratory analysis suggests—and what it cannot establish

INSIGHTS FROM THE DATA

1. Landing success generally increased over the years shown, but the rate did not rise monotonically.

2. Success rates differ across orbit categories; GTO is lower than several other categories in the displayed chart.

3. The SQL output contains differently formatted Success labels, showing why category normalization matters before drawing conclusions.

INTERPRETATION & LIMITATIONS

- These are descriptive patterns, not proof that year, orbit, or payload mass causes a landing outcome.

- Rare orbit categories may have very few launches, so a 100% observed rate may be unstable.

- All four classifiers show 83.33% accuracy in the supplied output; further metrics and validation are needed before claiming one model is better.

Conclusion & Next Steps

What the analysis demonstrates

CONCLUSIONS

- SQL and EDA helped summarize launch sites, outcomes, and trends.
- Yearly landing success generally improved, with fluctuations.
- Orbit and launch context show differing outcome patterns.
- Classification models reached 83.33% displayed test accuracy.

NEXT STEPS

- Normalize inconsistent categorical labels.
- Report precision, recall, F1, and confusion matrices consistently.
- Validate performance with cross-validation and a held-out test set.
- Improve the map’s zoom level and include clear distance / marker legends.
- Capture a genuine dashboard screenshot for the final submission.

Repository & Reproducibility

Project artifacts and final deliverable

GITHUB REPOSITORY

github.com/Yossefassem79/SpaceX-Falcon-9-Landing-Prediction

The repository includes notebooks for data collection, web scraping, data wrangling, SQL/EDA, machine learning, a Dash dashboard script, and the dashboard dataset.

Thank you